Lab 3: Component Modelling & Architectural Pattern Selection

PROBLEM STATEMENT #35 | Retail, E-Commerce & Finance

Customizable Subscription Box Scheduler

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Architecture Selection: We chose the Layered Architecture for the Customizable Subscription Box Scheduler System.

Architectural Choice:

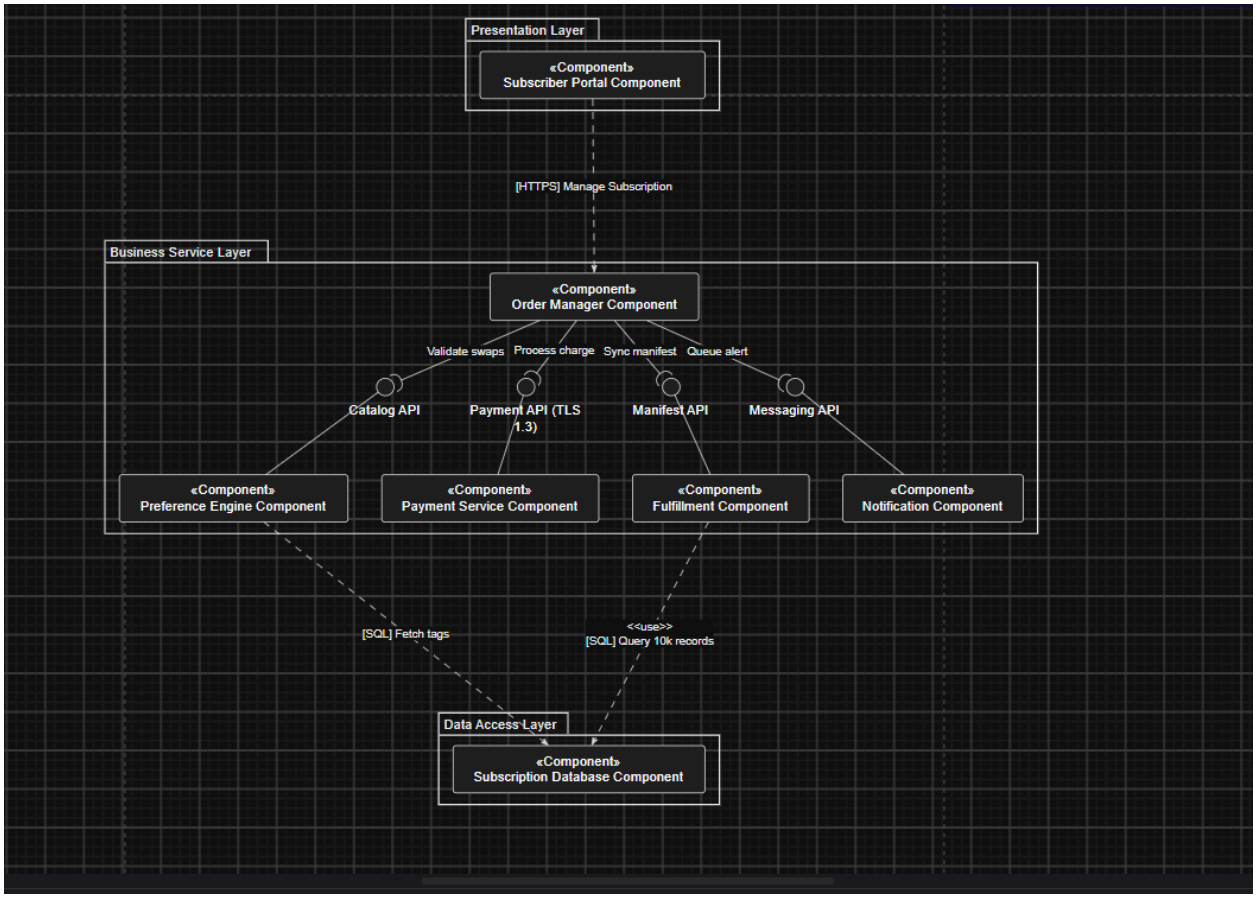
The selected architectural style is a 3-Tier Layered Architecture. In theory, this style organizes components into horizontal, hierarchical tiers where each layer only communicates with the layer directly below it. In a real-world enterprise context, this enforces a strict separation of concerns—isolating the user interface (Presentation), the core orchestration and business rules (Business Service), and the data persistence mechanisms (Data Access) into manageable, independent domains.

Two Reasons:

1. **Isolation of UI from Business Rules (FR-001 & FR-002):** The system demands complex logical checks, such as enforcing a strict 48-hour lockout window for swapping items or pausing deliveries. By utilizing a Layered Architecture, the Subscriber Portal Component (Presentation Layer) is decoupled from this logic. Frontend developers can iterate on the UI for configuring preference tags or swapping items without risking accidental modifications to the critical time-validation logic housed securely within the Order Manager Component.
2. **Centralized Data Management (FR-004):** The Fulfillment Lead requires a consolidated monthly manifest grouped by delivery zone and product type. A unified Data Access Layer simplifies the execution of these massive, complex table joins. Instead of scattering database credentials and querying logic across multiple services, the Data Access Layer centrally manages all connections, ensuring that the heavy SQL queries executed by the Fulfillment Component are tightly controlled and optimized.

Security Advantage: The Layered Architecture inherently defends against direct data exposure through strict boundary enforcement. The non-functional requirements mandate that subscriber payment profiles and personal preference data must be encrypted using AES-256 at rest and TLS 1.3 in transit. Because the Presentation Layer is physically and logically barred from communicating with the Data Access Layer, a compromise at the Subscriber Portal Component does not grant an attacker direct SQL execution rights. Every request must be routed through the Order Manager Component, acting as a gatekeeper that ensures authentication and authorization protocols are met before any encrypted payload is retrieved or modified.

Performance Benefit: This architecture optimizes resource utilization for high-volume database operations through centralized connection pooling. The system is required to handle high-volume seasonal spikes and export shipping labels for 10,000 monthly boxes in under 1 minute. Because all database interactions pass through a dedicated Data Access Layer, the system can leverage optimized database connection pools and query caching. When the Fulfillment Component triggers its massive data pull, the Data Access Layer can prioritize these read operations and manage I/O throughput highly efficiently, preventing the database server from being overwhelmed by disparate, uncoordinated connections.


Presentation Layer
«Component»
Subscriber Portal Component
[HTTPS] Manage Subscription
Business Service Layer
«Component»
Order Manager Component
Validate swaps
Process charge
Sync manifest
Queue alert
Catalog API
Payment API (TLS 1.3)
Manifest API
Messaging API
«Component»
Preference Engine Component
«Component»
Payment Service Component
«Component»
Fulfillment Component
«Component»
Notification Component
[SQL] Fetch tags
<<use>>
[SQL] Query 10k records
Data Access Layer
«Component»
Subscription Database Component